

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

Audience: Technology executives and network infrastructure leaders

17%

Traffic growth

YoY global Internet traffic

225

Disruptions

Major outages observed

6.5%

Mitigated

All traffic processed

13%

PQ TLS 1.3

Up from ~2%

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Global Internet traffic growth

Year-over-year increase in 2024

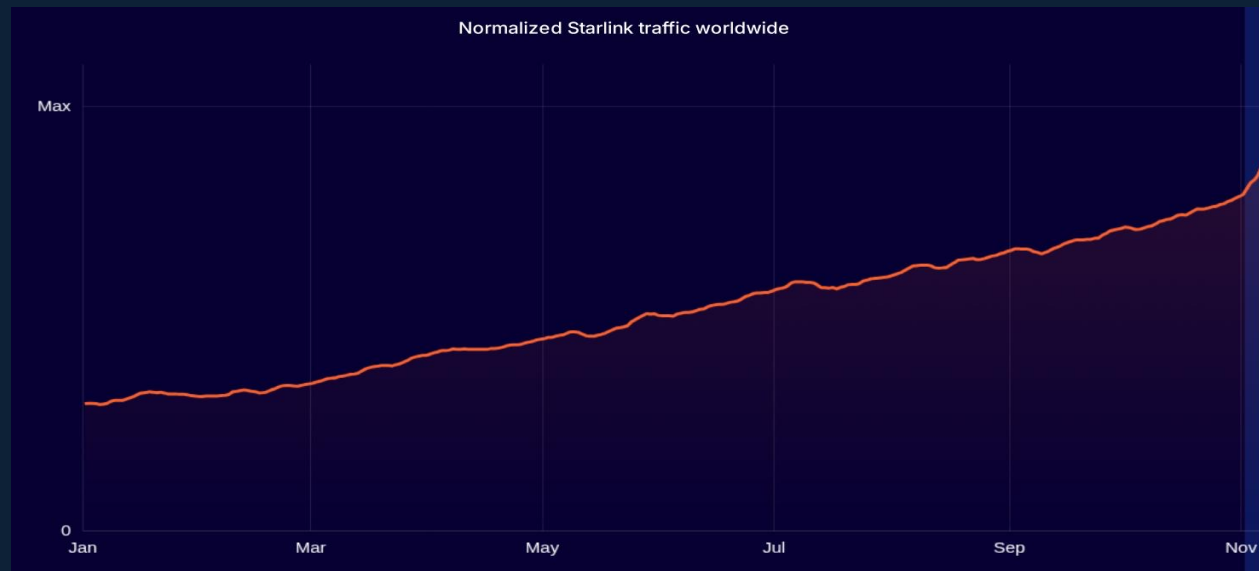
3.3x

Starlink traffic surge

Scaled normalized request volume

EXECUTIVE SIGNAL

Demand expansion continues across communication, commerce, entertainment, and transportation; satellite networks add reach in underserved regions.



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Google Leads Top 10 Services as Generative AI Rises

TOP 10 INTERNET SERVICES

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Google #1 for second consecutive year

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background. Ranking methodology is based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



Post-quantum encrypted traffic worldwide after Chrome 124 enabled key agreement by default on April 17

~2%

Early 2024 PQ share

TLS 1.3 traffic baseline

12–

13%

Rapid climb

Following Chrome 124 rollout

21%

HTTP/3 adoption

Runs on QUIC with encryption by default

~12%

Go API clients

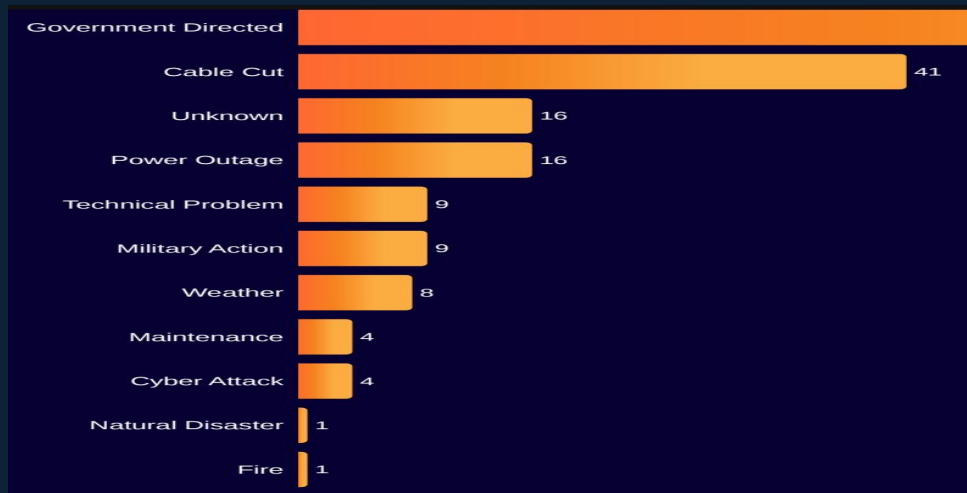
Top automated API request language

Other adoption markers: iOS generated 33% of traffic; Google held 89% of search referral traffic; Chrome remained the most popular browser globally.

225 Major Disruptions: Government Shutdowns Lead Outage Causes

CAUSE BREAKDOWN

Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1



Causes of major Internet disruptions observed in 2024

28%

IPv6 dual-stack traffic

41%

Global mobile traffic

Spain recorded the highest download speed among measured countries.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic mitigated

Share of all traffic processed

68%

Bot traffic concentration

Originated from top 10 countries

4.3%

Emails malicious

Among emails processed

43%

Deceptive links

Among malicious emails

+6.4%

RPKI valid routes

Routing security improvement

MOST-ATTACKED SECTOR

Gambling and Games

Cloudflare identified this industry category as the most attacked in 2024.

PERSISTENT VULNERABILITY

Log4j remains targeted

CVE-2021-44228 continues to be actively exploited more than two years after disclosure.

EMAIL THREAT MARKER

.bar

Top-level domain originating the largest share of malicious and spam email observed during the year.

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic growth trajectory

17% YoY growth confirms Internet dependency across communications, commerce, entertainment, and transportation.

2

Satellite reach expands

Starlink's 3.3x traffic surge signals continued expansion into underserved locations.

3

Post-quantum momentum

PQ encryption moved from ~2% to 13% of TLS 1.3 traffic after Chrome 124 default enablement.

4

Connectivity fragility

225 disruptions were observed, with government-directed shutdowns leading at 116+ incidents.

5

Persistent security pressure

6.5% traffic mitigation, Log4j targeting, and malicious email metrics keep defense modernization urgent.

2025 planning lens: prioritize scalable capacity, cryptographic agility, routing resilience, and threat-informed security operations.